RAJALAKSHMI ENGINEERING COLLEGE

Dharanipriyan S 251801056-2025-AIDS D2

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|---|---|
| **Started on** | Saturday, 12 September 2026, 5:42 PM |
| **State** | Finished |
| **Completed on** | Saturday, 12 September 2026, 5:43 PM |
| **Time taken** | 40 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n - Size of array

Next n lines Contains n numbers - Elements of an array

Last Line Contains Integer x - Sum Value

**Output Format**

First Line Contains Integer - Element1

Second Line Contains Integer - Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int a[], int l, int r, int x)
{
    if (l > r)
        return -1;

    int mid = (l + r) / 2;

    if (a[mid] == x)
        return mid;

    if (a[mid] > x)
        return binarySearch(a, l, mid - 1, x);

    return binarySearch(a, mid + 1, r, x);
}

int findPair(int a[], int l, int r, int n, int x)
{
    if (l > r)
        return 0;

    int need = x - a[l];

    int pos = binarySearch(a, l + 1, n - 1, need);

    if (pos != -1)
    {
        printf("%d\n", a[l]);
        printf("%d\n", a[pos]);
        return 1;
    }

    return findPair(a, l + 1, r, n, x);
}

int main()
{
    int n, i, x;

    scanf("%d", &n);

    int a[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

```

```
51    if (findPair(a, 0, n - 1, n, x) == 0)
52        printf("No");
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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